

Synthesis and Student Research Designs

Labor II — Week 13

Teaching deck draft

Central question and why a synthesis week matters

- How do firm behavior, frictions, institutions, and shocks fit into one labor-market framework?
- Week 13 is not a loose recap. It turns the course sequence into a research toolkit.
- The capstone discipline is to name the mechanism, unit, observed margin, and equilibrium scope before making a big claim.

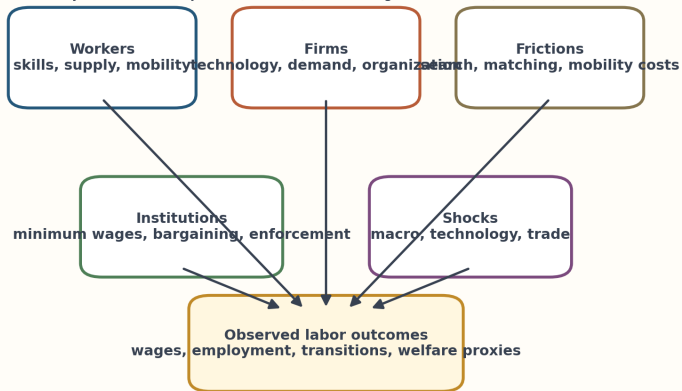
Labor I to Labor II bridge

- Labor I centered workers: labor supply, human capital, mobility, inequality, discrimination, and household constraints.
- Labor II embedded those worker-side objects inside firms, wage-setting, search frictions, institutions, and major shocks.
- Together they form one field: worker choices filtered through market structure and institutional environments.

Labor II architecture map

Labor II architecture: interacting objects, not separate silos

A capstone question always asks which object moves first, which margin is observed, and which channels remain offstage.



One architecture object for the semester

$$Y_{u,t+h} = \mathcal{G} \left(X_{u,t}^{\text{worker}}, X_{u,t}^{\text{firm}}, \mathcal{F}_{m,t}, \mathcal{I}_{m,t}, \mathcal{S}_{m,t:h} \right)$$

- The point is not that this is already a structural model.
- The point is to keep workers, firms, frictions, institutions, and shocks on the same conceptual page.
- The unit u changes with the question, and so does the claim the design can support.

Choosing units of analysis

- Worker: earnings, employment, mobility, welfare distribution
- Firm or establishment: labor demand, organization, compliance, adoption
- Occupation or task: exposure, job content, classification
- Region or place: local incidence, spillovers, reallocation
- Market or regime: institutional coverage and equilibrium comparison

Question → mechanism → unit → design

Good Labor II projects become sharper as each choice disciplines the next one



The most common failure is skipping from an interesting topic directly to a regression before the unit and observed margin are clear.

Descriptive, reduced-form, and structural approaches

$$\Delta y_{u,t+h} = \beta D_{u,t} + X'_{u,t} \Gamma + \varepsilon_{u,t+h}$$

- Descriptive work is strongest when the field lacks a clean object or measurement frame.
- Reduced-form work is strongest when treatment, mechanism, and observed margin line up tightly.
- Structural work is necessary when the question itself is about equilibrium propagation, welfare, or policy counterfactuals.

Comparative synthesis: wage-setting and labor market power

- Labor demand sets the value of labor to firms.
- Search, bargaining, and posting shape surplus division and wage dispersion.
- Monopsony turns mobility frictions into markdowns and firm labor-market power.
- Unions and wage regulation change voice, coverage, compression, and incidence.

Comparative synthesis: regulation and institutions

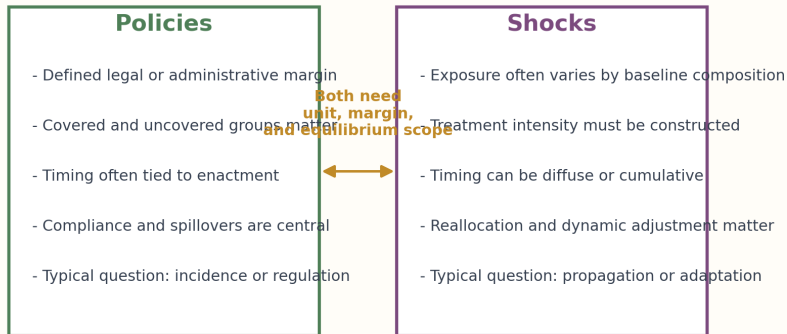
- Ask four questions: which margin is targeted, who is covered, where do spillovers go, and what welfare object is implied?
- Minimum wages, bargaining systems, labor law, enforcement, and insurance are distinct institutional treatments.
- Coverage without enforcement is weak treatment; direct effects without spillovers are incomplete interpretation.

Comparative synthesis: macro, technology, and trade adjustment

- Macro shocks: vacancies, unemployment, separations, wage cyclicalities
- Technology shocks: tasks, adoption, learning, local reallocation
- Trade shocks: import exposure, offshoring, place incidence, long-run worker adjustment
- Across all three, short-run incidence and long-run welfare are different objects.

Policies versus shocks

Policy evaluation and shock evaluation overlap, but they are not identical design problems



Partial equilibrium versus equilibrium reasoning

$$\Delta W \neq \omega \cdot \beta$$

- A wage, employment, or transition coefficient is not automatically a welfare statement.
- Place effects do not automatically equal worker welfare effects.
- Short-run treatment effects do not automatically reveal long-run adjustment.
- Stronger claims require explicit spillover, incidence, and aggregation structure.

How to build a research idea

- Question: what labor-market outcome needs explanation?
- Mechanism: through which disciplined channel?
- Unit: worker, firm, task, place, or market?
- Data: which margin is directly observed?
- Design: descriptive, reduced-form, or structural?
- Contribution: what does the project add beyond a re-description of an existing paper?

Common failure modes

Most weak capstone ideas fail before estimation begins

Vague mechanism

Interesting topic, but no disciplined

Mismatched unit

Teachments and observation un-

Hidden equilibrium claim

Observed, full welfare is implied.

Short run = long run

Immediate incidence is narrated as durable adjustment.

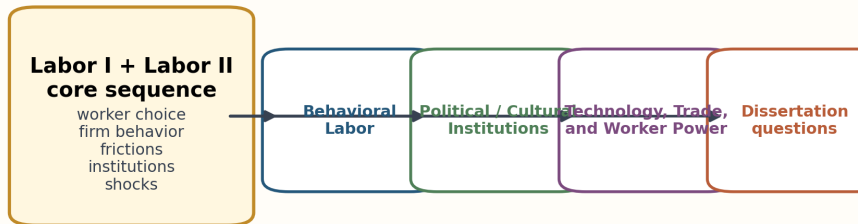
Design too big

The adjustment is really an unbounded dissertation.

Week 13 is strongest when it teaches students to name these failure modes early and fix them while the project is still small.

Frontier directions and bridge beyond Labor II

Week 13 should end by opening doors to the next projects, not by closing the course



Bridge beyond the sequence

- Back to Labor I: worker heterogeneity, inequality, mobility, and human capital with stronger market-side discipline
- Forward to Behavioral Labor: search behavior, worker beliefs, and application technology under frictions
- Forward to institutions: bargaining, norms, enforcement, and political or cultural structure as labor-market objects
- Forward to dissertation work: bounded questions with clear units and honest equilibrium scope

Takeaways

- Labor II works best as one architecture linking workers, firms, frictions, institutions, and shocks.
- Unit choice, observed margin, and equilibrium scope are core design choices, not afterthoughts.
- Descriptive, reduced-form, and structural approaches answer different questions and should not be ranked mechanically.
- The capstone goal is to turn a mechanism into a research design that is sharp, bounded, and useful.

Appendix: Week 13 lab bridge

- Diagnose: identify mechanism, treatment unit, observation unit, observed margin, and equilibrium concern.
- Compare: pair one anchor paper with a nearby design that changes the unit or treatment family.
- Design: produce a memo scaffold for a bounded extension rather than a vague final-project idea.
- The studio artifact is a disciplined starting point for field papers, workshops, or dissertation questions.